PHYC10009 Foundations of Physics

Light and Lenses

16

Professor Martin Sevior
School of Physics
University of Melbourne

THE UNIVERSITY OF MELBOURNE

Location:

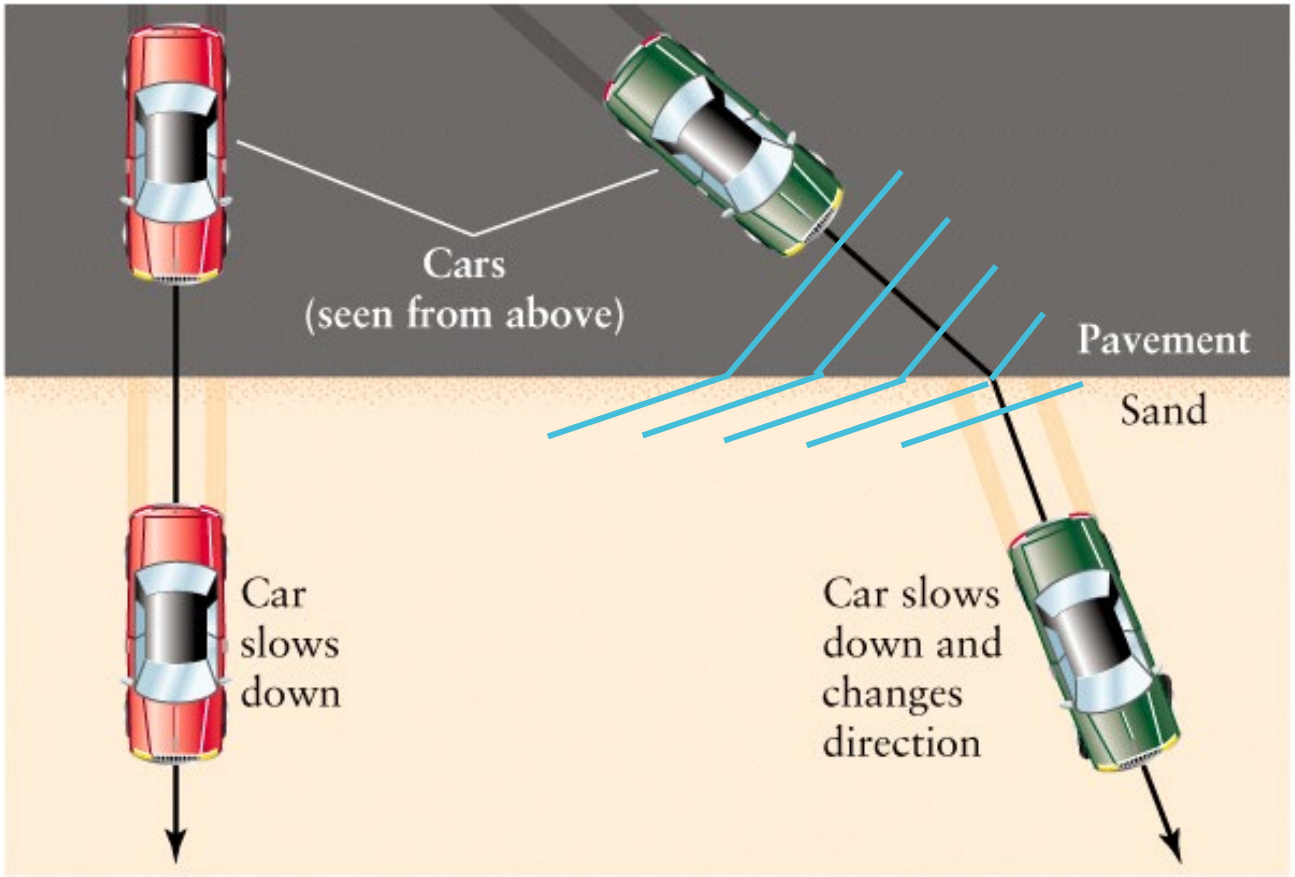
Room: 501 – Physics Building

Email: martines@unimelb.edu.au

REFRACTION: SNELL'S LAW

Refraction

- Reduced velocity of wavefront passing across boundary
- $v_2 < v_1$ so wave "bends" towards normal
- Consider car running off asphalt onto sand…



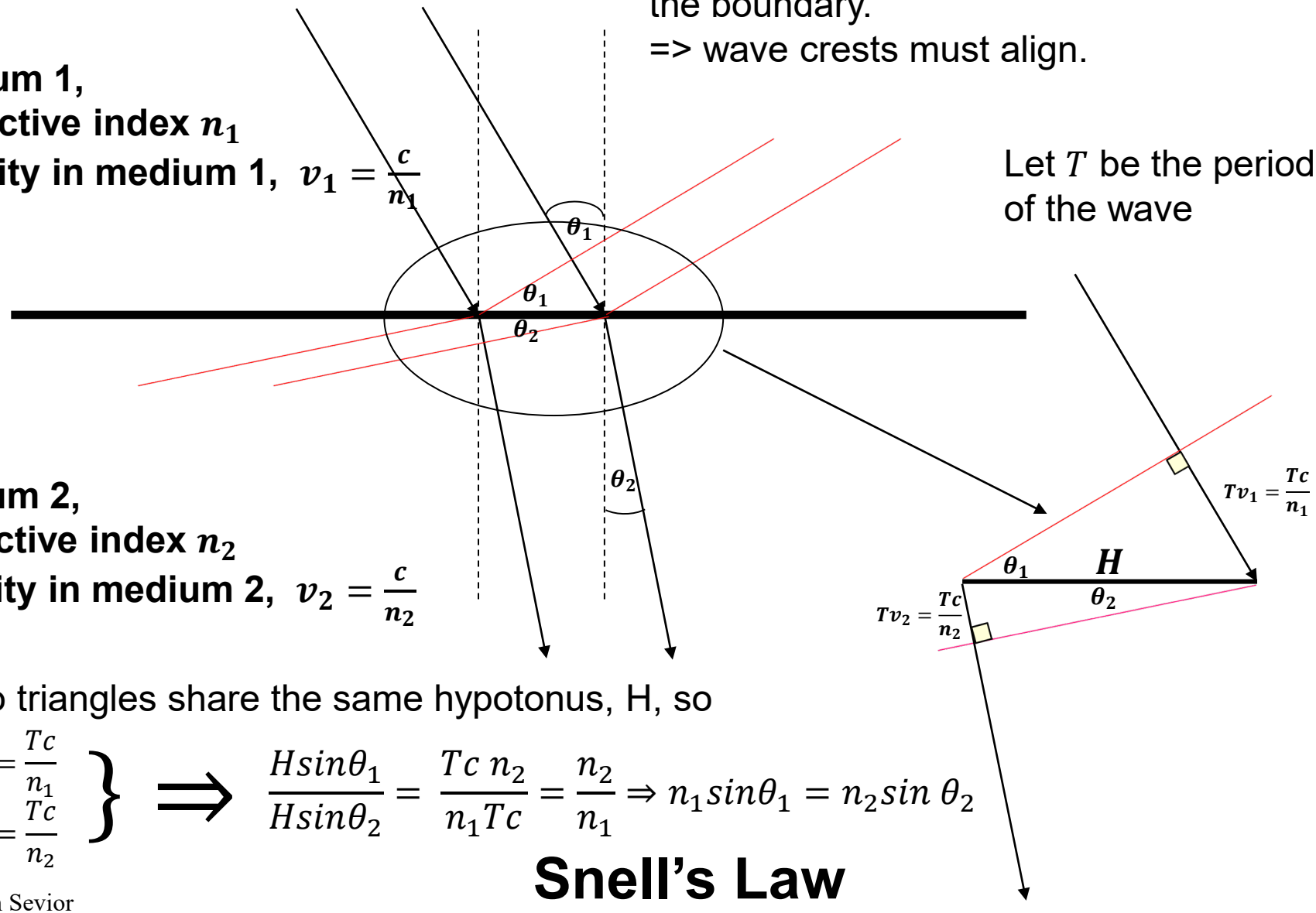
a

Snell's Law

Phase of wave in medium1 must match phase in medium 2 across the boundary.
=> wave crests must align.

Medium 1,
Refractive index n_1
velocity in medium 1, $v_1 = \frac{c}{n_1}$

Let T be the period of the wave



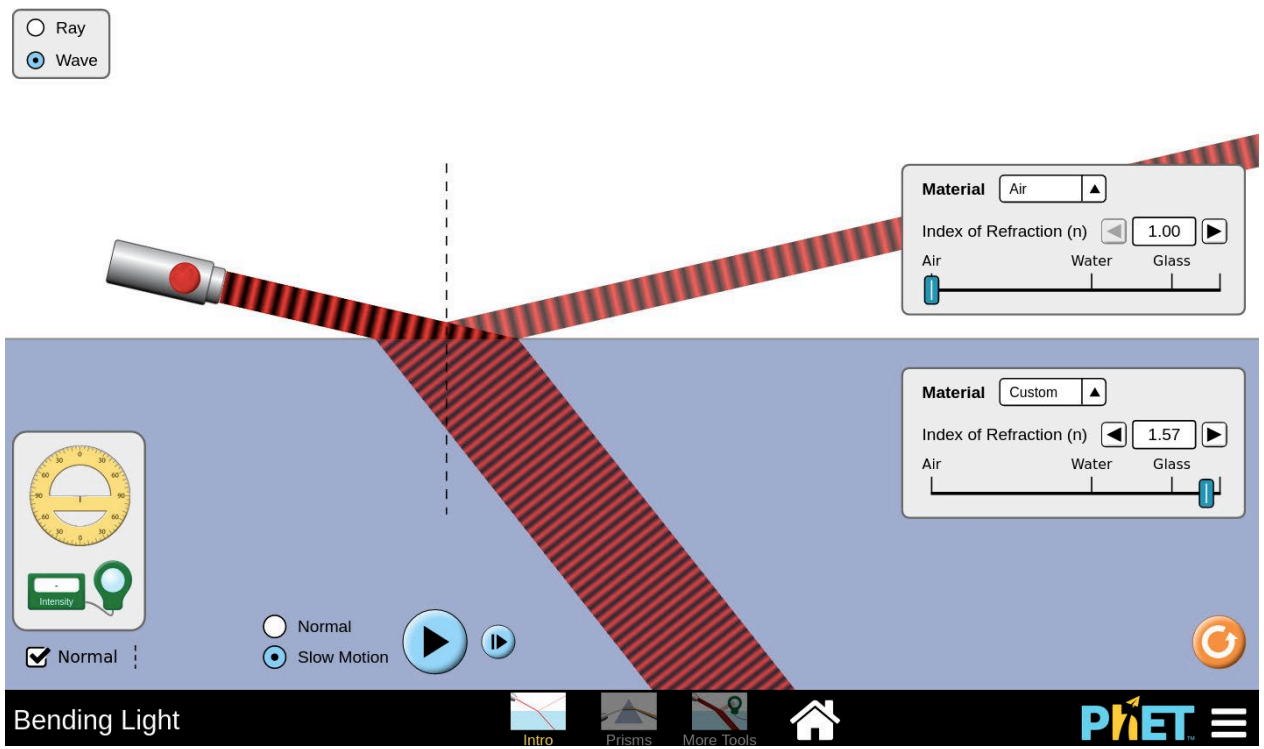
Medium 2,
Refractive index n_2
velocity in medium 2, $v_2 = \frac{c}{n_2}$

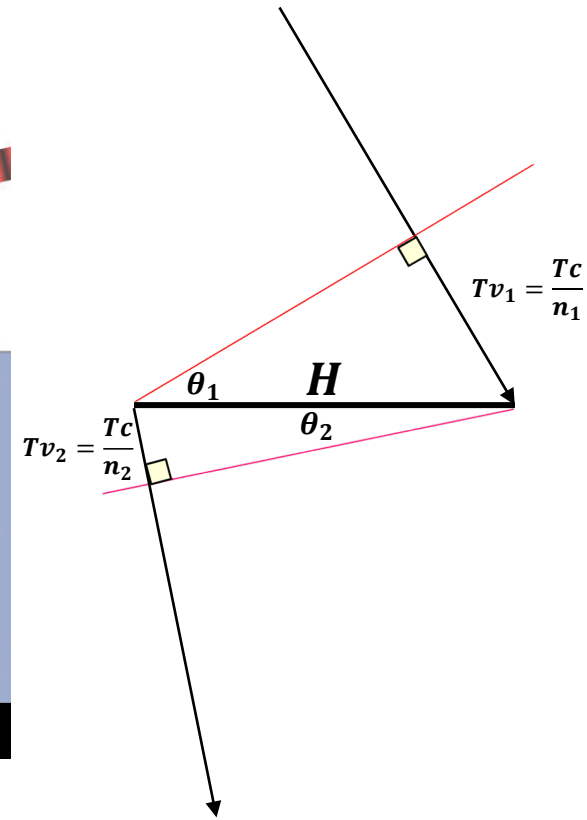
The two triangles share the same hypotonus, H, so

$$\left.\begin{aligned} H\sin\theta_1 &= \frac{Tc}{n_1} \\ H\sin\theta_2 &= \frac{Tc}{n_2} \end{aligned}\right\} \Rightarrow \frac{H\sin\theta_1}{H\sin\theta_2} = \frac{Tc\,n_2}{n_1 Tc} = \frac{n_2}{n_1} \Rightarrow n_1\sin\theta_1 = n_2\sin\theta_2$$

Snell's Law

Snell's Law Simulation





$$n_1 sin\theta_1 = n_2 sin\ \theta_2$$

https://phet.colorado.edu/sims/html/bending-light/latest/bending-light_en.html

Law of refraction (Snell's Law)

- Willebrord Snel van Royen (1580? – 1626)
  aka Willebrord Snellius

- Three important points:
  1. Ray changes direction (refracts) because leading wavefront slows
  2. Wavelength decreases because frequency unchanged and speed reduced
  3. Beam in the glass is wider, hence intensity is less inside

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

D Ob-3 Different Refractive indices

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Polar bear, Melbourne Zoo
Photo courtesy Shane Huntington

2008 exam question

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Question 11 **7 marks**

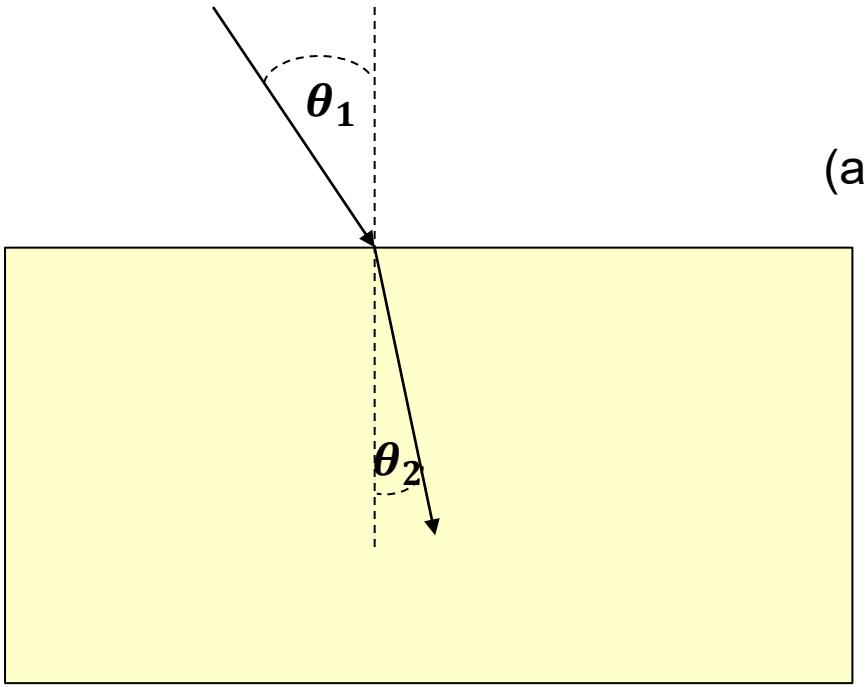
A beam of light from a laser in vacuum has a wavelength of 632.8 nm and is incident on the surface of a glass slab. In the vacuum the beam makes an angle of 32.0° with the normal to the surface, while in the glass it makes an angle of 21.0° with the normal.

(a) What is the index of refraction of the glass? (To help in answering this question, draw a diagram in your exam booklet to illustrate this situation.)

(b) What is the wavelength of the light in the glass?

[4 + 3 = 7 marks]

Solution



(a) What is the refractive index of the glass?

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

n_1 is in vacuum so $n_1 = 1.0$

$$n_2 = \frac{\sin(\theta_1)}{\sin(\theta_2)} = \frac{0.5299}{0.3584} = 1.48$$

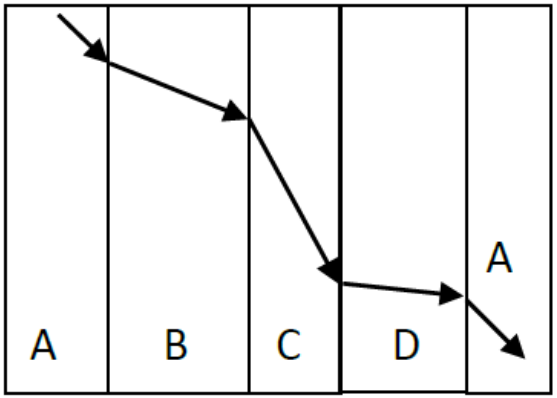
(b) What is wavelength of the light in the glass?

$\lambda_2 = \frac{n_1}{n_2} \times \lambda_1 = \frac{632.8}{1.48}$ = 427.6 nm

Example exam question: 2016

Question 8 **11 marks**

(a) The figure at right shows a ray of monochromatic light passing through several different materials, A, B, C and D and A again. Rank the materials according to their index of refraction, smallest first.

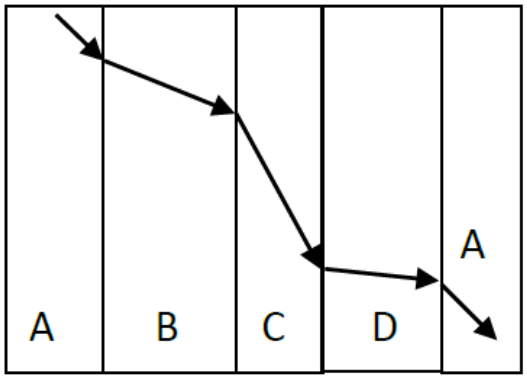


D Ob-1 Matching Refractive indices

Solution to exam question

Question 8 **11 marks**

(a) The figure at right shows a ray of monochromatic light passing through several different materials, A, B, C and D and A again. Rank the materials according to their index of refraction, smallest first.



Looking at refractive indices **A,B,C,D**
A -> B Light bends towards normal so **B** > **A**
B -> C light bends away from normal so **B** > **C**
Angle difference B -> C is greater than A -> B so **A** > **C**
C -> D light bends towards normal so **D** > **C**
D -> A light bends away from normal so **D** > **A**
Angle difference D -> A is greater than A -> B so **D** > **B**
Putting it all together gives:
C < **A** < **B** <**D**

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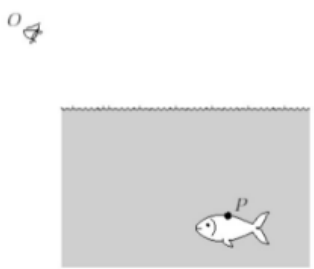
Depth of Fish

A fish swims below the surface of the water at *P*. An observer at *O* sees the fish at



1. Greater depth than reality

2. The same depth

3. Smaller depth than reality

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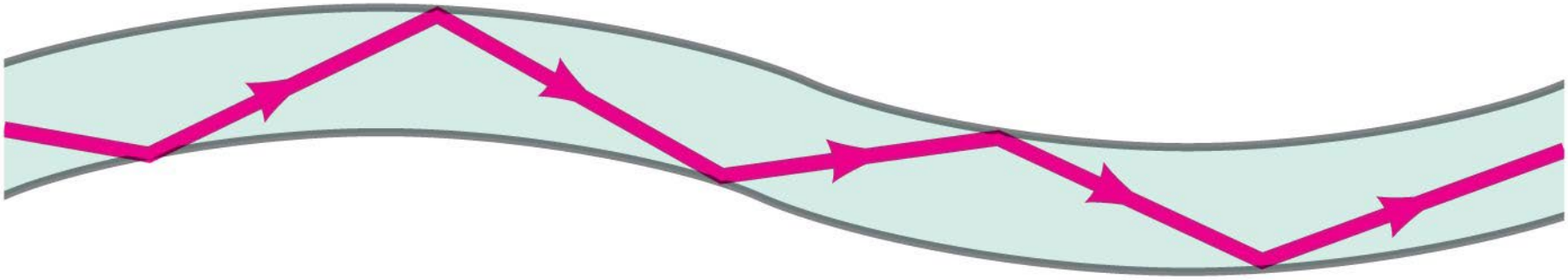
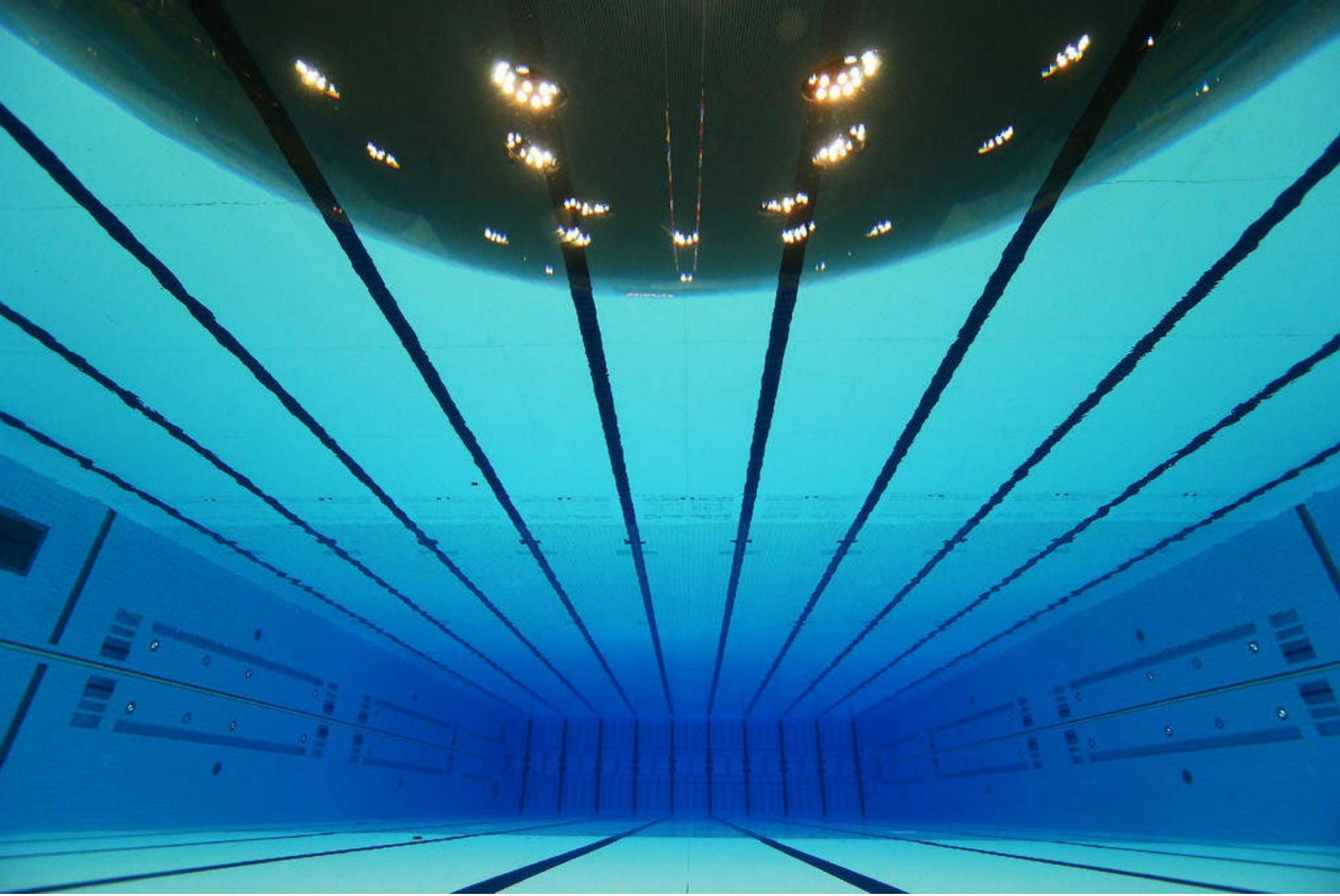
23.6 TOTAL INTERNAL REFLECTION





http://wordpress.mrreid.org/wp-content/uploads/2012/09/2012-poolcam-photograph.jpg

23.6 Total internal reflection

- From high to low refractive index: light bends *away* from normal
- We have $n_1 \sin\theta_1 = n_2 \sin\theta_2$ or…

$$\sin\theta_2 = \frac{n_1}{n_2}\sin\theta_1$$

- But what if $n_2 < n_1 \sin\theta_1$? θ_2 is not defined!

- *Total internal reflection – a perfect mirror if*:

$$\sin\theta_1 > \frac{n_2}{n_1} \quad \text{or} \quad \theta_1 > \sin^{-1}\frac{n_2}{n_1}$$

$$\theta_c = \sin^{-1}\frac{n_2}{n_1}$$

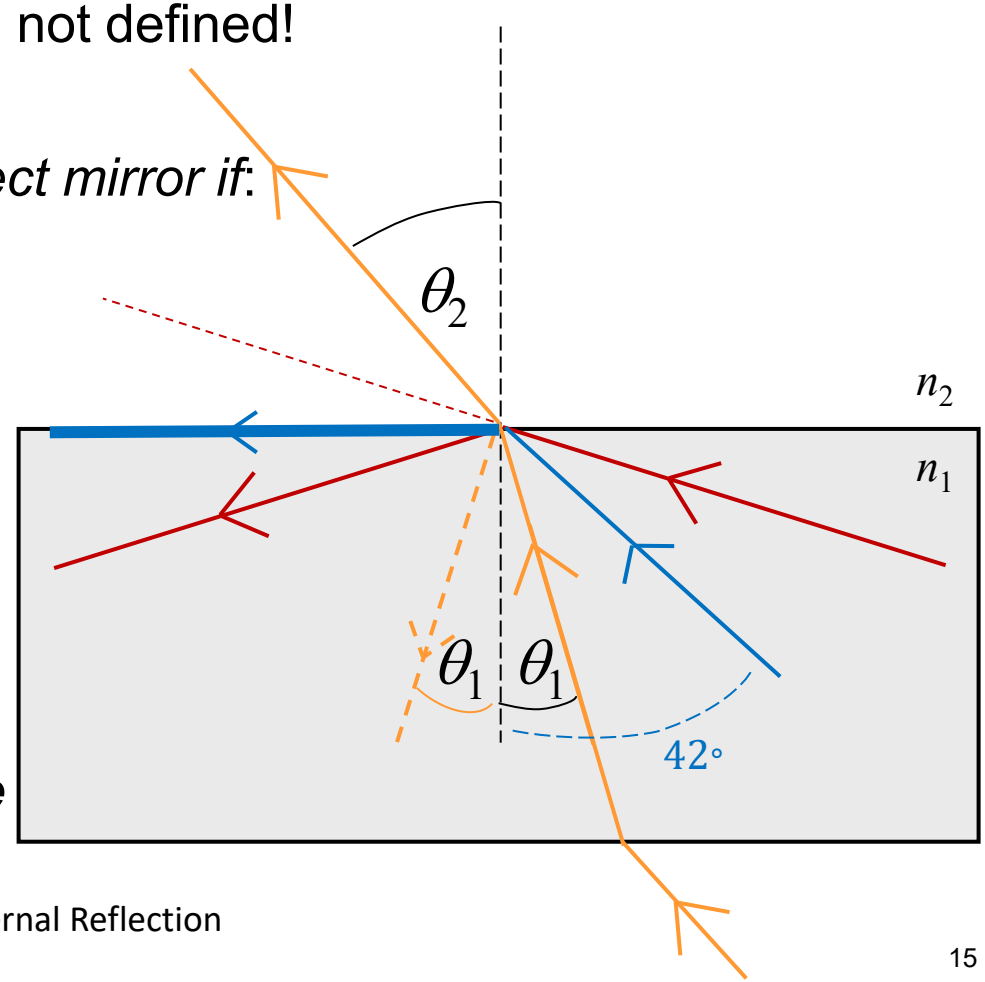
- Glass/air: $\frac{n_2}{n_1} = \frac{1}{1.5}$

  Critical angle $\theta_c = 42°$

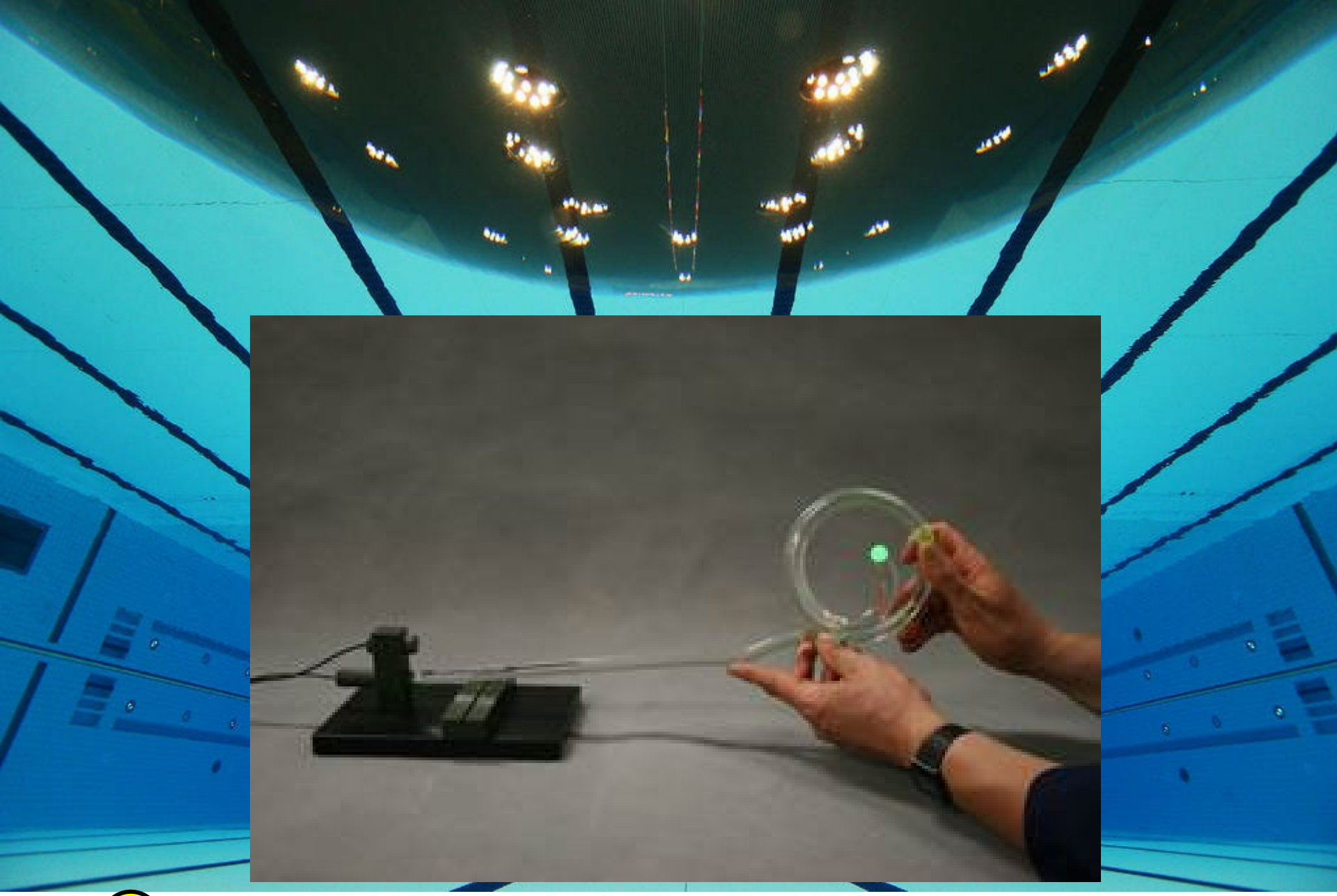
- Water/air: $\frac{n_2}{n_1} = \frac{1}{1.33}$ $\theta_c = 49°$

  Underwater: look up at surface

  – *Mirror-like*



D Oe-4 Total Internal Reflection



D Oe-1 Light pipe

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This lecture

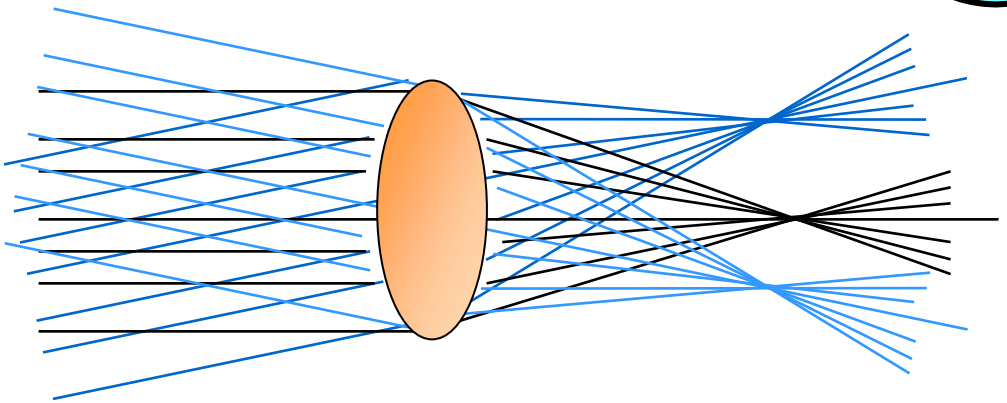
Imaging

- How lenses produce images
- The Thin Lens Equation and image formation
- The human eye

THIN LENS RAY TRACING

23.7 Thin lens ray tracing

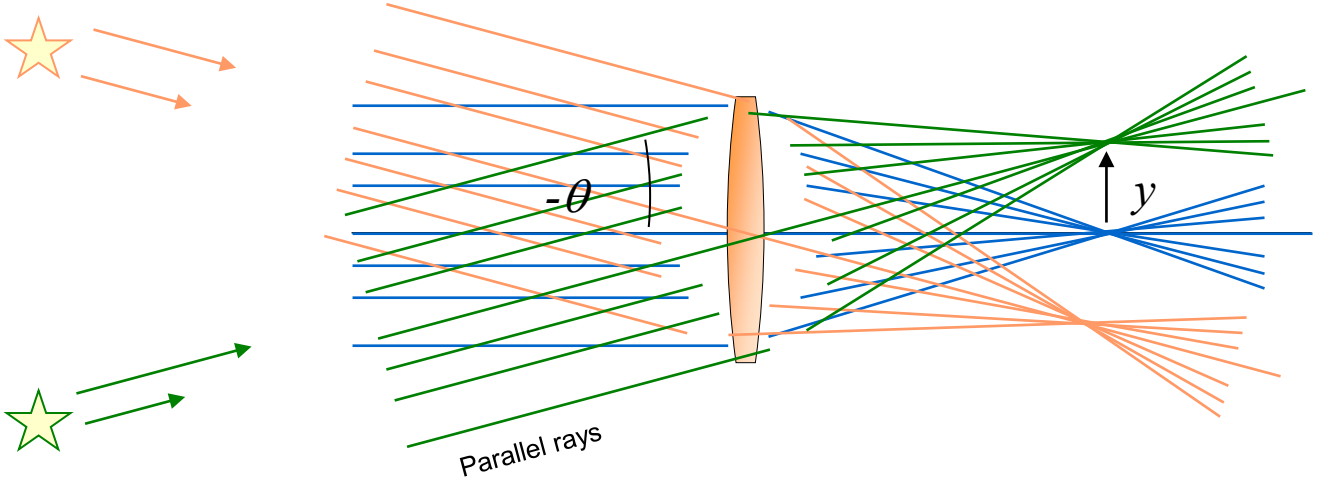
- Fundamental imaging device (e.g. our eyes depend on them!)
- What does a lens do?
- It collects and focuses light
  - e.g. melt ants with magnifying glass + sun's rays
- ***A lens converts the direction of rays to position***



- All rays coming from one direction pass through common point after lens – regardless of where they strike the lens
- This point is the ***focal point*** for a given direction
- Different directions → different distances from optic axis $y \approx -f\theta$

Lens $y = -f\theta$

- A lens transforms direction to position

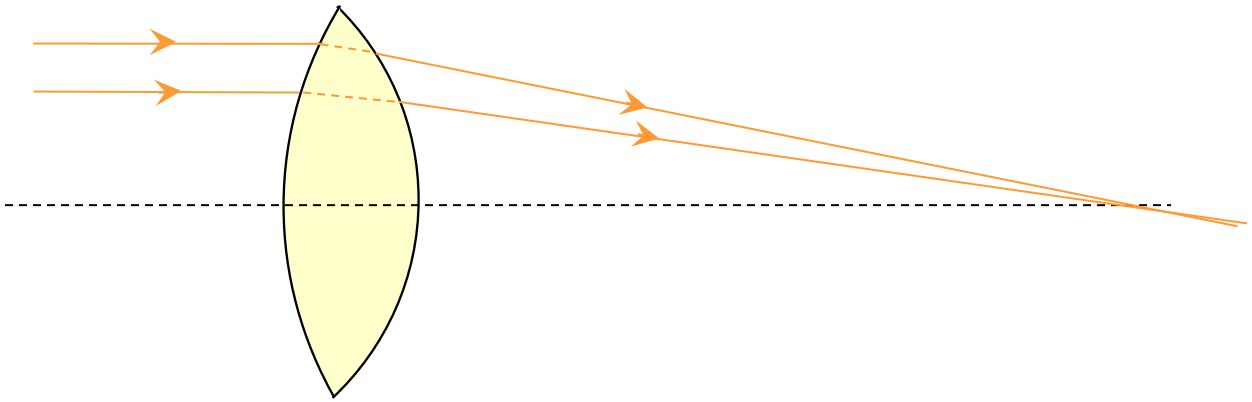


$$y = -f\theta$$

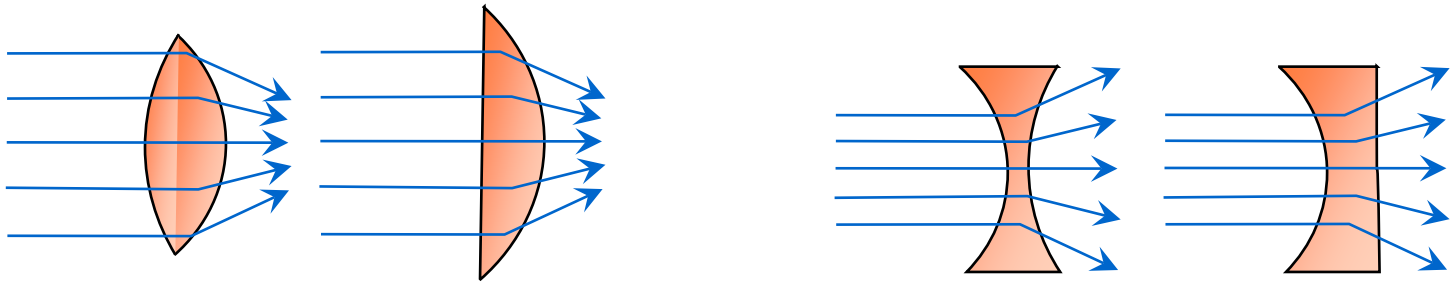
- Different directions → different distances from optic axis

Thin lenses

- Most lenses are ***thin***
- Rays leave the lens well before they focus



- Convex – converging – positive: thicker at the middle
- Concave – diverging – negative: thinner at the middle

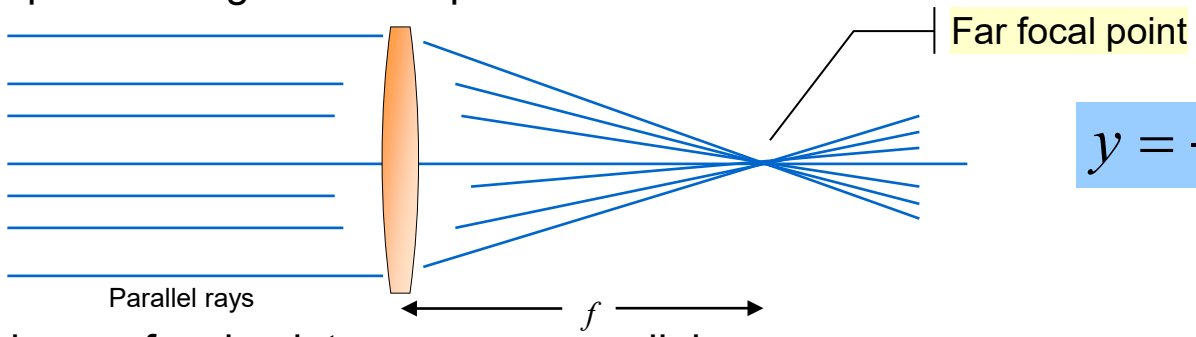


https://phet.colorado.edu/sims/html/bending-light/latest/bending-light_en.html

Thin lenses, ray tracing, three examples

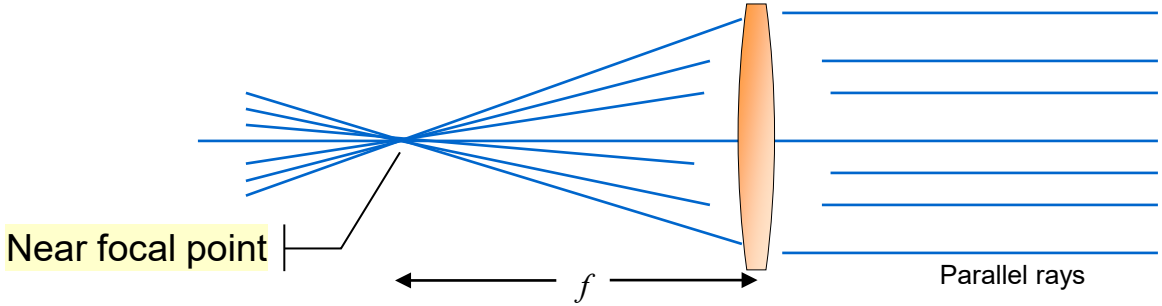
Three important examples for a thin converging lens

- Parallel rays pass through far focal point



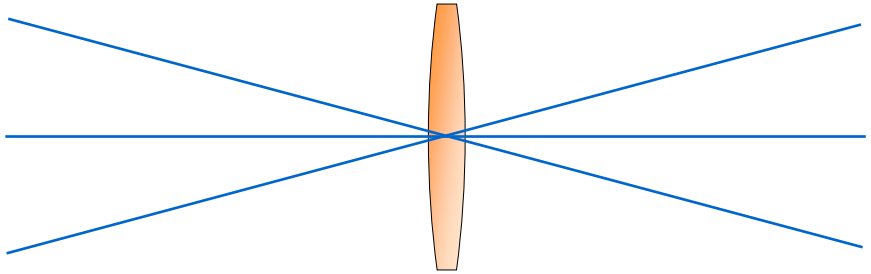
$$y = -f\theta = 0$$

- Rays through near focal point emerge as parallel rays



D Oa-3 Ray tracing

- Rays through lens centre are not bent



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The focal length of a lens is:

(a) the diameter of the lens

(b) the thickness of the lens

(c) the distance from the lens at which incoming parallel rays bend to intersect at a point

(d) the distance from the lens at which incoming parallel rays bend to intersect at a point

(e) the distance at which the lens should be placed from a point source of light, to create a parallel beam.

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